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Pivotality and responsibility attribution in sequential voting☆

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ABSTRACT

This paper analyzes responsibility attributions for outcomes of collective decision making processes. In particular, we ask if decision makers are blamed for being pivotal if they implement an unpopular outcome in a sequential voting process. We conduct an experimental voting game in which decision makers vote about the allocation of money between themselves and recipients without voting rights. We measure responsibility attributions for voting decisions by eliciting the monetary punishment that recipients assign to individual decision makers. We find that pivotal decision makers are punished significantly more for an unpopular voting outcome than non-pivotal decision makers. Our data also suggest that some voters avoid being pivotal by voting strategically in order to delegate the pivotal vote to subsequent decision makers.

"As soon as a majority has voted for it, it is declared passed, and the member who voted last is given credit for having passed it."—Shapley and Shubik[1]

1. Introduction

Many economic and political decisions are taken collectively. For example, boards commonly make decisions about business strategies in firms, committees of experts decide on interest rate policies in central banks, and coalition governments often enact laws in democratic countries. This paper analyzes how people affected by a collective decision, such as workers, shareholders, or electorates, attribute responsibly for the decision outcome to individual members of the collective decision making entity. Understanding responsibility attribution for collective decisions is of relevance because it can, for example, affect shareholders' willingness to extend a manager's contract or influence a political party's prospects of reelection.

Collective decisions are often reached by a vote among the decision makers and the voting process is often transparent. For example, the Bank of England's Monetary Policy Committee reveals its members' voting decisions and explicitly states that the "decision goes to the majority and there is no attempt to arrive at a consensus: members are individually accountable for their decisions" (see also Bank of England, 2005). The minutes of the U.S. Federal Reserve Open Market Committee are published as well (Levy, 2007).[2]

In this paper, we focus on analyzing responsibility attribution for outcomes of collective decisions reached by a transparent voting process. In particular, we ask if decision makers are blamed for being *pivotal*

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[1] Shapley and Shubik (1954), p. 788.

[2] Both the Monetary Policy Committee and the Federal Reserve Open Market Committee publish individual votes, and the latter also reveal the order of votes, namely: first chairman, than vice chairman, then the other members in alphabetical order (see also Gerlach-Kristen and Meade, 2011).

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if they implement an unpopular outcome in a *sequential voting process*. This question is of interest also in light of the above quotation from Shapley and Shubik (1954) who introduced the idea that the pivotal voter in a collective decision making process "is given credit," i.e., is held responsible for having passed the decision. Hence, our paper can also be seen as a test of their proposition.

We employ an incentivized laboratory experiment to address our research question. In the experiment, there are groups of six subjects. Three subjects have voting rights and decide sequentially and observably whether to implement an equal or an unequal allocation of money among themselves and the other three subjects, who have no voting rights. The equal allocation gives the same amount of money to all subjects, whereas the unequal allocation increases the monetary payoff of the subjects with voting rights at the expense of those without voting rights. A simple majority rule applies. After the vote, subjects without voting rights can assign costly punishment points to the voters. We interpret the assignment of punishment points as a measure of responsibility attribution.

Our main finding is that subjects attribute significantly more responsibility to the pivotal voter than to the other voters. The result holds even if we control for standard punishment motives such as outcome based fairness (e.g., Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000), unkind intentions (e.g., Rabin, 1993; Dufwenberg and Kirchsteiger, 2004), or the interaction of outcomes and intentions (Falk and Fischbacher, 2006). Our data further suggest that about one-fifth of those voters who reveal their preference for the unequal allocation when their vote as the last voter is decisive vote for the equal allocation if it is possible to "delegate" the pivotal vote to the subsequent voter.[3]

Our study is closely related to recent experimental work in political science by Duch et al. (2015), who also examine responsibility attributions for collective decisions. They consider a setting that, like ours, is akin to a collective dictator game with punishment. Their design, however, is different in two important ways. First, they consider a simultaneous voting procedure, while we focus on sequential voting. Second, one decision maker has proposal power and decision makers have weighted votes in their design, while the voters in our design are symmetric (apart from the sequential order of voting). Duch et al.'s main finding is that the decision maker with proposal power and the one with the largest voting share incur the most punishment in case of an unequal allocation. The sequential voting procedure in our experiment renders the role of pivotality more salient. Our work thus complements Duch et al.'s findings by revealing the importance of the pivotal vote for responsibility attribution in collective decision making.

Our paper also contributes to the political science literature that does not focus on collective decision making in particular but on responsibility attribution in general. In this literature, the attribution of credit or blame has been related to the power of a decision maker (see e.g. Banzhaf, 1964; Penrose, 1946; Shapley and Shubik, 1954), the number of veto players (Tsebelis, 2011) and governing party size (see e.g. Anderson, 1995; Lewis-Beck, 1990), and to the extent to which unified control of policymaking by incumbent governments is possible (Powell and Whitten, 1993). Similarly, Finer (1975), Alesina (1997), Lijphart (2012), and Franzese (2002) argue that coalition governments provide less potential for electoral accountability than single party governments, and Duch and Stevenson (2008) report that voters are more likely to attribute economic outcomes to single-party majority cabinets than to coalition governments.

Finally, the results of our study contribute to the economics literature on the importance of pivotality in markets and organizations (see e.g. Falk and Szech, 2013), as well as to the literature on delegation of unpopular decisions (e.g., Hamman et al., 2010; Coffman, 2011; Bartling and Fischbacher, 2012). Falk and Szech (2013) analyze how the decision maker's perception of her own pivotality affects the likelihood of taking an immoral decision (the decision to kill a mouse) in a trading environment. They find that the likelihood of killing the mouse is higher if a decision maker's perception of being pivotal is lower and that on the aggregate level, many more mice are killed in a treatment where pivotality is diffused. Our study shows that not only the perception of the pivotality of the own decision matters for choice. We show that more responsibility and blame are attributed to pivotal than to non-pivotal decision makers, which in turn can affect the decision makers' choices. Bartling and Fischbacher (2012) demonstrate that it is possible to shift the blame for an unpopular decision by delegating the choice to another person, and that many people do so. The main result of our current paper shows that it is also possible in the context of collective decision making to shift some blame by "delegating" the pivotal vote, and the data suggest that some voters make use of that option.

The remainder of our paper is organized as follows. Section 2 explains the experimental design and procedures. We discuss the punishment predictions of standard social preference models in economics, as well as the role of pivotality for punishment in Section 3. Section 4 reports our experimental results. Section 5 concludes.

2. Experimental design

We implemented a sequential voting game with punishment. Three "voters" and three "receivers" form a group. The voters decide on the allocation of a total of 30 points among the six group members, using a simple majority rule. There are two possible allocations. The unequal allocation gives 9 points to each of the voters and only 1 point to each of the receivers (9,9,9;1,1,1). The equal allocation distributes the 30 points evenly among the six group members (5,5,5;5,5,5). Importantly, the voters cast their votes sequentially. The other voters and receivers of the group are able to observe both the sequence of the decisions and the decisions themselves. First, Voter 1 votes for one of the two allocations. Then Voter 2 observes Voter 1's action and votes herself. Finally, Voter 3 casts her vote, knowing the choices of Voters 1 and 2. Abstentions are not possible. Fig. 1 illustrates the decision tree.

The three receivers first observe the sequence of the votes and thus also the voting outcome. One randomly selected receiver then has the option to punish individual voters by deducting points. Punishing is costly for the receivers. A receiver incurs a fixed cost of one point to be able to deduct up to seven points from the voters. The seven punishment points can be assigned to a single voter or they can be distributed among two or all three voters, but it is not possible to reduce a voter's payoff below zero.

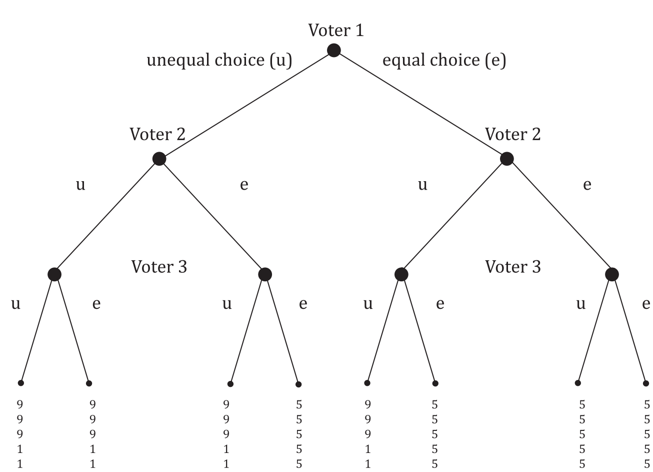

Fig. 1. Voters' choices and resulting allocations for voters and receivers.

[3] A vote by the third voter is decisive if the first two voters fail to reach a majority.

The players' payoff functions are summarized as follows. A voter either receives nine or five points, depending on the chosen allocation; furthermore, he might incur punishment points from the randomly chosen receiver. The randomly chosen receiver gets either one or five points, depending on the chosen allocation, minus the cost of punishment (one point) if she deducts at least one point from the voters. The two other receivers get a payoff of either five or one point, depending on the chosen allocation.

The game was played one-shot and we used the strategy method for both receivers and voters. Each receiver decided, in a randomized order, for each of the eight possible voting histories (see the eight end-nodes in Fig. 1) how to punish the individual voters. Only after the receivers made all eight decisions, each receiver learned whether or not she was the randomly selected receiver who could punish, and she knew that her punishment decisions for the realized situation were binding.[4] Voters decided at seven decision nodes, namely as first as Voter 1, then as Voter 2, and finally as Voter 3. In the role of Voters 2 and 3, subjects make decisions for every possible pre-play history (see the seven decision nodes in Fig. 1). The respective pre-play histories were shown in a randomized order. Voters knew that their choices were binding and that they would learn their randomly assigned role as Voter 1, Voter 2, or Voter 3 only after they decided on all seven decision nodes.[5]

2.1. General procedures

Before the subjects entered the lab, they randomly drew a place card that specified at which computer terminal to sit. The terminal number determined both a subject's role (voter or receiver) and the group matching. After entering the lab, subjects received paper copies of the instructions explaining the game at their assigned terminals. The subjects' instructions included comprehension questions that had to be answered correctly before a session began. We used a neutral framing for voters and receivers (Players A and B), as well as for punishment (deduction points), and for the labeling of the two allocations (Allocations 1 and 2). An experimenter read a summary of the instructions aloud to ensure common knowledge about the game. An English translation of the original German instructions can be found in the Online Appendix. The experimental data and programs used for data collection are available on this journal's webpage.

The data were collected in four sessions in two consecutive weeks in November and December 2012. 144 subjects participated in total. Each subject participated only once. The sessions took place at the decision laboratory of the Department of Economics at the University of Zurich. Participants were students from the University of Zurich and the Swiss Federal Institute of Technology in Zurich. The experiment was conducted using z-Tree (Fischbacher, 2007), and we used ORSEE (Greiner, 2004) for recruiting.

The experiment took about 75 min. Each experimental point was converted to CHF 3.00 at the end of the experiment. The subjects earned about CHF 26.30 (about $28 at the time of the experiment), which included a show-up fee of CHF 12. Earnings were paid privately to ensure the anonymity of the decisions.

[4] Brandts and Charness (2011) provide a survey of the effect of the use of the strategy method on punishment decisions in experiments and find no case where a treatment effect that is detected using the strategy method vanishes when the direct response method is used. The use of the strategy method can thus be considered as the more conservative test in our context.

[5] Although we used the strategy method, we made the sequential voting procedure very clear. For each of the possible situations, each voting decision was reflected by an individual update of the computer screen. For example, the third voter was informed about the preceding voters' decisions in the sequential order and had to confirm the receipt of each piece of information by clicking a button. Equivalently, a receiver had to click through a sequence of screens showing the sequential order of the three voters' decisions (i.e., the path through the game tree) before making her punishment decision for a given voting outcome. See the experimental instructions in the Online Appendix for more details.

3. Punishment motives

If all subjects are purely self-interested, the receivers will not incur the cost of punishment, irrespective of the resulting allocation and the voting sequence. It is well known, however, that the pure self-interest model does not always accurately predict many subjects' behavior. In this section, we discuss punishment motives based on the most widely used economic theories of social preferences. We include punishment predictions of social preference models based on outcomes (e.g. Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000), intentions (Rabin, 1993; Dufwenberg and Kirchsteiger, 2004), and the interaction of outcomes and intentions (Falk and Fischbacher, 2006). In addition, we consider “choice” (i.e., vote) as one particular punishment motive because it strikes us as a natural heuristic to assign punishment in our context. Finally—being the focus of this paper—we discuss the potential punishment motive “pivotality.”[6]

3.1. Outcome-based models of social preferences

Outcome based models of social preferences, e.g., Fehr and Schmidt (1999) or Bolton and Ockenfels (2000), assume that some people dislike unequal allocations. Inequity-averse receivers might thus be willing to incur the cost of punishment to reduce payoff differences. This model class predicts no punishment if the equal allocation prevails but predicts positive punishment otherwise. Importantly, the predicted punishment neither depends on individual choices (votes) nor on the sequence of voting.

3.2. Choice of the unequal allocation

Voting for the unequal allocation could be perceived as a stated preference for the unequal allocation and thus as blameworthy. This would predict no punishment for a voter who opted for the equal allocation but some punishment for a voter who opted for the unequal allocation (even if this voter can no longer influence the outcome). The punishment motive “unequal choice” can be seen as a naïve version of the punishment motive “intentionality,” which we discuss next.

3.3. Intention-based models of social preferences

Intention-based models of social preferences, e.g., Rabin (1993) or Dufwenberg and Kirchsteiger (2004), assume that people are willing to incur costs to punish unkind actions, irrespective of the resulting allocation. In these models, the unkindness of an action is measured by comparing the chosen action with the possible alternative actions. In our game, voting for the unequal allocation is an intentionally unkind action if a voter is still able to affect the outcome of the voting process. If Voters 1 and 2 already decided the vote, Voter 3's vote is irrelevant and therefore classified as a neutral action, i.e., as neither kind nor unkind. Voting for the equal allocation, while still being able to affect the outcome, is a kind action. The notion of intention-based reciprocity predicts no punishment for neutral or kind voters.

3.4. Outcome and intention

Falk and Fischbacher (2006) combine the punishment motives “outcome” and “intention.” In their model, people are willing to punish decision makers who implement unequal allocations, and punishment is

[6] The predictions of the formal economic models relate to predictions by concepts included in psychological attribution theories, such as the concept of “controllability” (Weiner, 1979), i.e., the extent to which a person was in control of causing a particular outcome. In the Online Appendix, we provide three conceptualizations of Weiner's idea of controllability and analyze the extent to which these conceptualizations explain the punishment decisions observed in our data. In line with the concept of controllability, we find higher punishment if an unfair action is chosen in a situation with higher control. We thank an anonymous referee for suggesting this analysis.

Table 1
Average punishment in the eight possible voting sequences.

Allocation	Voting sequence	Average punishment			Pivotal vs. non-pivotal	
		Voter 1	Voter 2	Voter 3	All voters	Unkind voter
Unequal	u-u-u	1.5	**1.85**	0.86	$p < 0.001$	$p = 0.294$
	u-u-e	1.86	**1.92**	0.26	$p < 0.001$	$p = 0.960$
	u-e-u	1.68	0.07	**2.39**	$p < 0.001$	$p = 0.006$
	e-u-u	0.11	1.83	**2.33**	$p < 0.001$	$p = 0.012$
Equal	u-e-e	1.33	0.10	0.08	–	–
	e-u-e	0.17	1.43	0.08	–	–
	e-e-u	0.06	0.03	0.92	–	–
	e-e-e	0.08	0.07	0.03	–	–

Notes: "u" denotes a vote for the unequal allocation; "e" denotes a vote for the equal allocation. The punishment for the pivotal voter is written in boldface. The two rightmost columns show *p*-values of Wilcoxon signed rank tests comparing the punishment for the pivotal voter to the punishment for the two other voters ("all voters") and to the punishment for the other intentionally unkind voter only ("unkind voter").

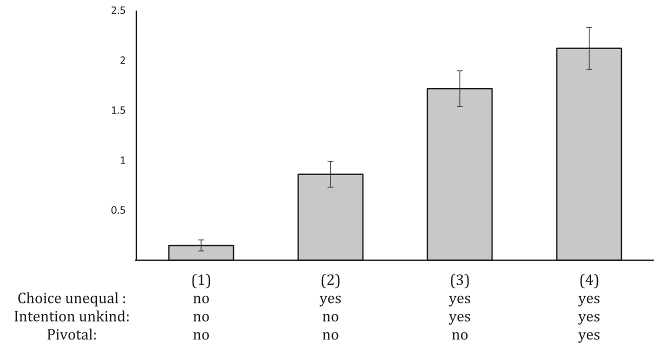

Fig. 2. Average punishment for voters if the unequal allocation results. The bars show the average punishment for (1) voters who voted for the equal allocation, (2) voters who voted for the unequal allocation but could not affect the outcome as it was decided already, (3) non-pivotal voters who intentionally voted for the unequal allocation, and (4) pivotal voters. The error bars show standard errors of the means.

even stronger if the decision maker's action is intentional. In our context, punishment for a voter is thus predicted only if both the unequal allocation results and the voter's intention is unkind.

3.5. Pivotality

The central hypothesis in our paper is that being pivotal is regarded as carrying special responsibility for the resulting voting outcome. Therefore, the prediction of the notion of pivotality is that the pivotal voter will be punished more than the other voters, given the unequal outcome results.

4. Results

4.1. Pivotality and punishment

Our main research question is whether receivers blame the pivotal voter more than the non-pivotal voters if the unequal allocation is chosen. We investigate this question using the punishment pattern as a measure of the assignment of blame. Table 1 shows the average punishment levels for the first, second, and third voter separately for each of the eight possible voting outcomes. The table shows that average punishment is higher for the pivotal voter (shown in boldface) than for the non-pivotal voters in all four voting sequences in which the unequal allocation results. For example, the first row of Table 1 shows the voting sequence "u-u-u," in which all voters opted for the unequal allocation. Voter 2 is the pivotal voter and she is punished by 1.85 points on average. Voter 1, the first intentionally unkind voter, is punished by 1.5 points and Voter 3, whose vote could not make a difference, is punished by 0.86 points on average. The two rightmost columns in Table 1 show the significance levels for the comparisons between the punishment for the pivotal voter and the non-pivotal voters, either taking the average of both other voters as the comparison group ("all voters") or the other intentionally unkind voter only ("unkind voter"). The table shows that the difference in punishment between the pivotal and the average of the two non-pivotal players is significant in all four voting sequences. While average punishment for the pivotal voter is higher than for the non-pivotal unkind voter in all four voting sequences, the differences in average punishment between the pivotal voter and the intentionally unkind voter are significant only in the two situations in which the pivotal voter is the last voter.[7]

In Fig. 2, we show the average punishment for all voters when the unequal outcome occurred. We aggregate punishment based on whether a voter's choice was unequal ("Choice unequal"), whether a voter was revealed to be intentionally unkind, i.e., the voter opts for the unequal outcome while the voting outcome is still open ("Intention unkind"), or whether the voter was pivotal, i.e., completed the minimum winning coalition necessary to implement the unequal outcome ("Pivotal"). Note that a voter's intention can only be unkind if her choice is unequal, and a voter can only be pivotal for the unequal allocation if she is intentionally unkind. We thus have four categories of voters, as shown in Fig. 2.

(1): Voters who did not choose the unequal allocation
(2): Voters who chose the unequal allocation but could no longer affect the outcome
(3): Voters who chose the unequal allocation when they could still affect the outcome, so that their action is intentionally unkind, but who were not pivotal
(4): Voters who chose the unequal allocation when they could still affect the outcome and who were pivotal.

Table 2 summarizes how the voters are placed into these four categories in the four different voting sequences that result in the unequal outcome.

Fig. 2 shows that average punishment is lowest for the voters who vote for the equal allocation. It amounts to 0.15 points and is highly significantly different from the other three groups (Wilcoxon signed rank tests based on average punishment per subject, $N = 72$, *p-values* < 0.001). The voters who chose the unequal allocation but could no longer affect the outcome (because Voters 1 and 2 opted for the unequal allocation already) were punished by 0.86 points on

Table 2
Voting sequences and placement of voters into categories in Fig. 2.

Voting sequence	Voter 1	Voter 2	Voter 3
u-u-u	(3)	(4)	(2)
u-u-e	(3)	(4)	(1)
u-e-u	(3)	(1)	(4)
e-u-u	(1)	(3)	(4)

Notes: The table shows which voters are placed into the four categories (1)–(4) shown in Fig. 2 for the different voting sequences that result in the unequal allocation. For example, the top row shows the voting sequence in which all voters voted for the unequal allocation (u-u-u). Voter 1 chose the unequal allocation, could still affect the outcome so that the unequal choice is intentionally unkind, but was not pivotal. Voter 1 thus falls into category (3) in Fig. 2. Voter 2 chose the unequal allocation, was intentionally unkind and pivotal. Voter 2 thus falls into category (4). Voter 3 chose the unequal allocation but could no longer affect the outcome, so she was neither intentionally unkind nor pivotal. Voter 3 thus falls into category (2).

[7] Throughout the paper, we only consider being pivotal *for the unequal allocation* and thus neglect the fact that the second voter who opts for the equal allocation after the first voter opted for the equal allocation is pivotal *for the equal allocation*. Little punishment occurs in case of the equal allocation, and we do not find that voters who are pivotal for the equal allocation are again punished significantly less than the other voters.

Table 3
An econometric comparison of different punishment motives.

OLS	(1)	(2)	(3)	(4)	(5)	(6)
Outcome unequal	1.024***					0.048
	(0.126)					(0.070)
Choice unequal		1.564***				0.782***
		(0.154)				(0.137)
Intention unkind			1.604***			0.517**
			(0.161)			(0.196)
Outcome unequal				1.565***		0.289
× Intention unkind				(0.169)		(0.232)
Pivotal					1.494***	0.403**
					(0.180)	(0.155)
Constant	0.365***	0.095***	0.208***	0.355***	0.628***	0.083**
	(0.067)	(0.033)	(0.041)	(0.055)	(0.065)	(0.037)
Observations	1728	1728	1728	1728	1728	1728
R^2	0.105	0.244	0.250	0.217	0.124	0.281

Notes: The dependent variable is punishment points for voters. *Outcome unequal* is a dummy variable which equals 1 if the unequal allocation is chosen. *Choice unequal* is a dummy variable which equals 1 if a voter opts for the unequal allocation. *Intention unkind* is a dummy variable which equals 1 if a voter opts for the unequal allocation and no majority was reached before her vote. *Pivotal* is a dummy variable which equals 1 if a voter is pivotal for the unequal allocation. Standard errors, clustered on 72 individuals, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

average, which is also highly significantly different from the other groups (Wilcoxon signed rank tests based on average punishment per subject, $N = 72$, *p-values* < 0.001). Finally, non-pivotal but intentionally unkind voters are punished on average by 1.72 points, while pivotal voters are punished by 2.12 points on average. This difference, as well, is statistically significant (Wilcoxon signed rank test based on average punishment per subject, $N = 72$, $p = 0.032$). We summarize our findings in the following:

Result 1. Pivotality and punishment

On average, the pivotal voter is punished the most.

4.2. An econometric comparison of different punishment motives

In this subsection, we provide an econometric comparison of the punishment motives that we discussed in Section 3. We first compare the explanatory power of the different motives in isolation. Then, we consider all motives simultaneously to analyze if pivotality has explanatory power on top of the other punishment motives.

We show the results of the econometric analysis in Table 3. Note that all decisions are always included in this analysis, including the decision resulting in the equal outcome. Regression (1) shows that the punishment for voters is significantly higher if the unequal allocation results (average punishment of 1.389 points) compared to if the equal allocation results (average punishment of 0.365 points). Regression (2) shows that those voters who vote for the unequal allocation are punished significantly more (average punishment of 1.659 points) than those voters who vote for the equal allocation (average punishment of 0.095 points). Regression (3) reveals that a voter with an unkind intention is punished significantly more (average punishment of 1.812 points) than a voter with a neutral or kind intention (average punishment of 0.208 points). Regression (4) shows that the punishment for intentionally unkind voters whose votes contributed to the final implementation of the unequal allocation (average punishment of 1.920 points) is significantly higher compared to all other voters (average punishment of 0.355 points). Finally, regression (5) shows that punishment for voters who are pivotal for the unequal allocation (average punishment of 2.122 points) is significantly higher than the punishment for all other voters (0.628 points).

The main insight provided by the regression analysis in Table 3 is that all five punishment motives that we discussed in Section 3 make the qualitatively correct comparative-static prediction with respect to average punishment levels. To compare the explanatory power of the different motives, we can directly compare the R^2 in models 1 to 5. This is meaningful because all of the models have just one parameter. We find that the predictive power of the punishment motive "pivotal" ($R^2 = 0.124$ in model 5) is higher than the predictive power of the motive "outcome unequal" ($R^2 = 0.105$ in model 1). But the predictive power of the motives that take "choice" or "intention" into account is higher, and it is highest for the motive "intention unkind" ($R^2 = 0.250$ in model 3).

Finally, we analyze whether the punishment motive "pivotal" has an explanatory power *on top of* the other motives by combining all five punishment motives in one regression. Regression (6) in Table 3 reveals that the motives "choice unequal," "intention unkind," and "pivotal" all contribute to the explanation of the punishment pattern. In particular, the dummy variable "pivotal" remains significant, showing that being pivotal matters for the assignment of blame even when we control for other punishment motives. We summarize our results in the following:

Result 2. Pivotality vs. standard punishment motives

The punishment motive "pivotality" has a significant explanatory power for average punishment levels even if we control for the standard motives "outcome unequal," "choice unequal," "intention unkind," and the interaction "outcome unequal × intention unkind."

4.3. Voting behavior

In this section, we report on the voters' behavior. Overall, we find that the unequal allocation results in 67.4% of the cases. Table 4 provides a detailed overview of the voting behavior of Voters 1, 2, and 3 at each decision node and the expected payoffs that resulted from the respective choices.

The top two rows of Table 4 show that 58.3% of all Voters 1 voted for "unequal" and thus 41.7% for "equal."[8] The table further reveals that the decision between voting for "unequal" and "equal" involves a tradeoff. Voting for "unequal" increases the probability of the unequal allocation and thus the expected number of points before punishment, whereas voting for "equal" decreases expected punishment. The three rightmost columns report the expected number of allocated points before punishment, the expected punishment, and the expected payoff (allocated points minus punishment) of a particular voting choice at a given decision node. For Voters 1, the gain from increasing the likelihood of implementing the unequal allocation by voting for "unequal" (8.53–6.53 = 2) outweighs the loss from increased punishment (1.70–

[8] The decisions of a subset of 43 voters, who voted for the unequal allocation as decisive Voters 3, are shown in the column to the right of the decisions of all voters. We discuss this subset later in this section.

Table 4
Voting decisions and expected payoffs at each decision node.

Voter	Voting sequence (decision in bold)	Voter decision [%]		Expected points before punishment	Expected punishment	Expected payoff
		All voters ($n = 72$)	Decisive Voters 3 voting for "unequal" ($n = 43$)			
1	**u**-...	58.3	76.7	8.53	1.70	6.83
	e-...	41.7	23.3	6.53	0.11	6.42
2	u-**u**-...	59.7	79.1	9	1.90	7.10
	u-**e**-...	40.3	20.9	7.83	0.08	7.75
	e-**u**-...	61.1	81.4	7.5	1.68	5.82
	e-**e**-...	38.9	18.6	5	0.07	4.93
3	u-u-**u**	22.2	18.6	9	0.86	8.14
	u-u-**e**	77.8	81.4	9	0.26	8.74
	u-e-**u**	70.8	100	9	2.39	6.61
	u-e-**e**	29.2	0	5	0.08	4.92
	e-u-**u**	62.5	100	9	2.33	6.67
	e-u-**e**	37.5	0	5	0.08	4.92
	e-e-**u**	1.4	2.3	5	0.92	4.08
	e-e-**e**	98.6	97.7	5	0.03	4.97

Notes: "u" denotes a vote for the unequal allocation; "e" denotes a vote for the equal allocation.

$0.11 = 1.59$), such that the expected payoff of voting for "unequal" (6.83) is larger than the expected payoff of voting for "equal" (6.42).

Turning to Voters 2, Table 4 shows that 59.7 and 61.1% of all Voters 2 voted for "unequal" subsequent to Voter 1 voting for "unequal" and "equal," respectively. Consider first the situation in which Voter 1 voted for "unequal." Voters 2 could then increase their expected payoff by voting for "equal." Voting for "unequal" in that situation yields 9 points before punishment and an expected punishment of 1.9 points. Voting for "equal" reduces the expected number of points before punishment (7.83) but it also reduces the expected punishment (0.08). The reduction in punishment ($1.9–0.08 = 1.82$) thus more than compensates for the reduction of the expected number of points before punishment ($9–7.83 = 1.17$). Consider now the situation in which Voter 1 voted for "equal." The expected payoff for Voters 2 is now higher when they vote for "unequal." In expectation, voting for "unequal" in this situation yields 7.5 points before punishment, a punishment of 1.68 points, and thus a payoff of 5.82 points. By voting for "equal," Voters 2 implement the equal allocation, i.e., a payoff of 5 points before punishment. The expected payoff is thus lower, even though there is almost no punishment (0.07) for Voters 2 in that case.

We finally turn to Voters 3. Consider first the two cases where Voters 1 and 2 opted for different allocations, such that Voter 3 has the decisive vote. In these cases, the expected payoff of Voters 3 is higher if they implement the unequal allocation. The increase in the payoff before punishment ($9–5 = 4$) is larger than the increase in expected punishment at the two respective decision nodes ($2.39–0.08 = 2.31$ and $2.33–0.08 = 2.25$). In both cases, we find that the majority of Voters 3 voted for "unequal" (70.8 and 62.5%). In the two cases where Voters 1 and 2 already implemented the unequal or equal allocation, Voters 3 maximize their expected payoffs by voting for "equal" because this reduces the expected punishment. We find that all but one Voter 3 voted for "equal" when Voters 1 and 2 implemented the equal allocation. However, 22% of Voters 3 (16 of 72) voted for "unequal" when Voters 1 and 2 implemented the unequal allocation. This latter finding is surprising, also in light of the low-cost theory of expressive voting (see e.g. Brennan and Lomasky, 1993), which would predict that of those Voters 3 whose votes are inconsequential, all opt for the socially more desirable equal allocation. Our findings on the behavior of these Voters 3 are more consistent with experimental results by Tyran (2004), who finds that many voters "bandwagon," i.e., vote for the option that they expect (or know) the majority of other voters to vote for.

A final question arises as a result of our main finding that pivotality increases the blame for the unequal allocation: Do those voters who reveal their preference for the unequal allocation avoid being pivotal for the unequal allocation? In our game, in case Voter 1 opts for the unequal allocation, do some Voters 2 "delegate" the pivotal vote by strategically opting for the equal allocation?

Recall that we used the strategy method for voters. This allows us to address the question above, as we can compare voters' decisions across decision nodes. In particular, we are interested in those voters who opt for the unequal allocation as decisive Voter 3 (i.e., in situations in which Voters 1 and 2 opted for different allocations). Observing a voter's choices at these decision nodes provides us with a measure of her preference over the two allocations. Note that a voter is even pivotal (and thus blamed most) when she votes for the unequal allocation at these decision nodes. Among our 72 voters, 43 always chose the unequal allocation as Voter 3 when their choice was decisive. [9] The respective column in Table 4 shows the voting behavior of these 43 voters separately. We find that 20.9% of these voters (9 of 43) opted for the equal allocation as Voter 2 subsequent to Voter 1 voting for the unequal allocation. This means that they delegate the pivotal choice to the subsequent voter—potentially in the hope that the last voter will secure the unequal allocation and take the blame.[10] We summarize this last observation in the following:

Result 3. Pivotality and voting behavior

About one-fifth of those voters who reveal their preference for the unequal allocation when their vote as the last voter is decisive avoid being pivotal for the unequal allocation as the second voter and delegate the pivotal choice to the subsequent voter.

5. Conclusion

In this paper, we addressed the question of responsibility attribution for individual members of collective decisions making entities, such as committees or boards. In particular, we analyzed whether people are blamed for being pivotal if they implement an unpopular outcome in

[9] Voter 3's choice is decisive at two decision nodes. We consider only those 43 voters who revealed twice and thus consistently that they prefer the unequal allocation at these decision nodes. Ten voters opted only once for the unequal allocation in the role of the decisive Voter 3. If we took them into account as well, the fraction of voters who delegate would slightly increase to 23% (12 out of 53).

[10] Table 4 further shows that the 43 voters whom we classify as having a preference for the unfair allocation based on their decisive vote as Voter 3, also vote more frequently than the other voters for "unequal" in the roles of Voter 1 and 2. 23.3% of the 43 voters vote for "equal" as Voter 1, a behavior that is consistent with the idea that these voters prefer to delegate the vote for the unequal allocation to subsequent voters. 18.6% of the 43 voters vote for "equal" in the role of Voter 2 if Voter 1 voted for "equal." This behavior could be interpreted as an aversion for being the first voter to vote for "unequal."

a sequential voting process. We measured responsibility attribution by assigned punishment points in an experimental voting game. Our main result is that pivotal decision makers are blamed significantly more than non-pivotal decision makers. This finding lends support to Shapley and Shubik's (1954) assumption that the pivotal voter in a collective decision "is given credit" for having passed the decision. Our results contribute to a better understanding of responsibility attribution for collective decision making in general and they have specific implications for the theoretical work on committee decisions, as they question, for example, the equivalence of transparent sequential and simultaneous voting procedures (see e.g. Levy, 2007).

Our experiment also showed that about one-fifth of voters who revealed a preference for the unequal allocation in the situation where their vote as the last voter is decisive opt for the equal allocation as the second voter subsequent to the first voter voting for the unequal allocation. This observation suggests that some voters strategically delegate the pivotal decision to subsequent voters. Our study thus extends existing experimental results by, e.g., Hamman et al. (2010), Coffman (2011), and Bartling and Fischbacher (2012), who show that many subjects shirk the responsibility for unpopular decisions by delegating the decision right.

The strength, and maybe the weakness, of our experimental design is that it reduces the decision-making context to one that renders pivotality very salient. While this helps isolate the effect of pivotality on responsibility attribution, the voting that actually takes place in collective decision-making entities outside the laboratory has a richer set of characteristics than our sequential, simple majority voting game over two given outcomes. For instance, recent results by Duch et al. (2015) indicate that other factors, such as proposal power and voting shares, are also important for responsibility attribution.

Moreover, our study focuses on a sequential voting game, whereas many voting processes exist where votes are cast simultaneously. However, our results on the effect of pivotality on responsibility attribution might be relevant also for simultaneous voting processes—in particular if decision makers' preferences over proposals are known. In the context of our game, consider, for example, a situation where it is publicly known that voter "U" strongly favors the unequal allocation and that voter "E" strongly favors the equal allocation. There is also a voter "P", but the only public knowledge is that she has neither a strong preference for the equal allocation (like voter type E) nor for the unequal allocation (like voter type U). Suppose further a committee of three voters that consists of one U, one E, and one P. Now, even with a secret and simultaneous vote, it is clear that P is the pivotal voter who tips the scales. Suppose the unequal allocation results. Then, on the one hand, U might be considered to be the most blameworthy "type" as it is known that she has the strongest preference for the unequal allocation. While the voting outcome reveals that P tends more towards the unequal than towards the equal allocation, she is nevertheless not type U, and thus potentially a less blameworthy "type." Hence, on the one hand, behavioral economics models of "type-based" reciprocity (e.g. Levine, 1998) would predict higher punishment for U than for P. Our results, on the other hand, predict that P would be punished for being pivotal, which could, in principle, lead to higher punishment for P than for U.

One natural interpretation of the above situation is a left/centrist or right/centrist coalition in parliament, possibly with a small centrist party and larger "anchor parties" on the left and the right, each not large enough to pass legislation on their own. How would constituents hold the coalition parties accountable for bad performance? Our result on the impact of pivotality on responsibility attribution would suggest that the centrist party, which tips the scales, would be held responsible. But alternative responsibility attribution heuristics that are discussed in the political science literature, such as proposal power or voting weights (Duch et al., 2015), would suggest that the respective larger anchor party would be held responsible as it typically also has proposal power. The example shows that it will be interesting to further explore in future research how people assign responsibility in collective decision making contexts, and how this depends on the specific features of the voting rules and the nature of the information available.

Appendix A. Supplementary material

Supplementary material to this article can be found online at http://dx.doi.org/10.1016/j.jpubeco.2015.03.010.

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